

Punit Label Application Flow and Label Format Guide

Google Docs-ready guide generated from the Flutter source on 22 June 2026.

1. Application Flow

1. Splash and login: the app starts at SplashView, then routes to LoginView when credentials are required. Login collects Company Code, email, and password, then stores authenticated user and token data.
2. Dashboard: DashBoardView loads user permissions, dashboard totals, top products, low-stock products, company details, device reconnection, and printer status polling. Role flags control whether Inward and Dispatch cards are accessible.
3. Global drawer configuration: the drawer controls default non-batch label format, white-label mode, serial number printing, timestamp printing, label copy count, tare behavior, printer/receipt mode, tower light status, scale connection, and printer connection.
4. Inward entry: operators select Batch Inward or Non Batch Wise Inward. Both flows capture gross, tare, and net weights, create barcodes from serial numbers, print labels, maintain logs, and submit pause/stop payloads to the API.
5. Dispatch: operators select a customer, scan or manually enter a barcode, verify and add dispatch items, then save the dispatch transaction and export reports.

2. Configuration That Changes Labels

Setting	Effect
White Label	Removes company header/footer from generic and special previews when enabled.
Serial Number	Prepends Sr No to labelFields where the selected label format supports it.
Time Stamp	Adds date/time to static previews and print payloads.
Label Copies	Repeats the platform print call from 1 to 10 times.
Tare State: OFF	Tare is forced to 0 and labels usually print Net Weight only.
Tare State: ON	Gross Weight, Tare Weight, and Net Weight can be printed.
Tare State: Barcode	Tare products can be selected/scanned and included in weighing logic.
Label vs Receipt	Label mode calls Android label-printer methods; Receipt mode sends item lines to the Bluetooth receipt printer.

3. Inward Workflow

Batch Wise

- Select a batch, then a product inside that batch.
- Select an existing label format. Batch flow currently previews existing/static formats.
- The controller syncs product tare unless global tare is OFF.
- Auto-weight mode monitors the scale every second. When the configured weight condition is met for the configured number of seconds, addToList creates a barcode, inserts a log entry, and prints the selected label.
- Pause/Stop groups barcodes by batch product and submits gross/tare/net/serial payloads, then generates an inward PDF report.

Non Batch Wise

- Enter a transaction name, select product, select label format, and choose product attributes.
- When custom label templates are supported, the UI adds Custom, label size, and custom option selectors.
- The app groups repeated entries by same product and same selected attributes; serial numbers continue inside that group.
- Existing/static labels build labelFields from selected attributes plus weight fields. Custom templates resolve field positions and values from the runtime template payload.
- Pause/Stop submits the transaction payload and generates a transaction PDF report.

4. Dispatch Workflow

- Open Dispatch from the dashboard when dispatch permission is enabled.
- Select a customer, then scan a barcode or type it into the Barcode field.
- verifyAndAddBarcode validates the barcode against dispatch data and appends it to dispatch logs.
- Save submits scanned barcodes. Export helpers can generate grouped PDF/Excel dispatch reports.

5. Label Format Catalog

Format	Preview/print size	Selectable field limit	Behavior
Majedar Tea	600 x 410	1	First non-weight attribute becomes the description; net weight prints as Weight; Code128 barcode.
Small	600 x 410	3	Generic label. Non-white mode allows up to 3

			selectable fields; preview switches to grid when more than 1 field is present.
Medium	600 x 600	4 or 6	Generic square label. Grid mode starts after 5 fields; white-label mode raises selectable fields to 6.
Large	700 x 600	5 or 8	Generic larger label. Default non-batch fallback is Large. Grid mode starts after 5 fields.
Extra Large	700 x 700	7 or 9	Generic large square label. Grid mode starts after 5 fields.
Wholesale Pack	700 x 1200	10	Tall generic label with Wholesale Pack title, business-hours line, product name, selected attributes, Gross Weight, and barcode.
Small Seven	600 x 410	5	Special two-column label. Excludes Weight and Sr No from attributes; unit conversion can insert Units.
DryFruit	600 x 410	3	Special dry-fruit layout. Excludes gross/tare/weight/sr from attribute list; weight fields are formatted to 3 decimals with Kg.
Custom Runtime	50x75, 75x75, 75x100, 100x100	Template driven	Fields come from API template coordinates: company, product, gross/tare/net, barcode, sr_no, datetime, footer, and attr_* fields.

6. Label Field Rules

- Small labels use Weight/Net Weight depending on the controller path; Medium, Large, and Extra Large usually include selected attributes plus Net Weight or Gross/Tare/Net depending on tare setting.

- Wholesale Pack keeps selected attributes and Gross Weight; Sr No is preserved when enabled.
- Majedar Tea ignores Sr No and weight keys when choosing its visible attribute description.
- DryFruit and SmallSeven remove weight-like keys from their attribute lists, then add their own weight/unit presentation.
- Runtime custom templates resolve field values by field_key. Supported built-in keys include company_name, company_email, company_contact_no, company_gst_no, company_website, company_address, product_name, gross_weight, tare_weight, net_weight, barcode, barcode_text, sr_no, datetime, footer, and attr_*

7. Label Preview Examples

The previews below use sample product data and recreate the app preview logic: label dimensions, visible fields, special filtering, grid thresholds, timestamp placement, and barcode placement.

Majedar Tea Label Format

Majedar Tea Co.

22-06-2026 14:35

Premium Assam Tea

Grade : BOP

Weight : 1.250



Small Label Select Max (3)

Punit Foods Pvt. Ltd.

22-06-2026
14:35

ops@punitfoods.in | +91 98765 43210
GST: 24AABCP1234Z1ZV

Besan Flour

Weight: 1.250



Label generated from weighing ERP by punitinstrument.com

Medium Label Select Max (4/6)

Punit Foods Pvt. Ltd.

22-06-2026
14:35

ops@punitfoods.in | +91 98765 43210

GST: 24AABCP1234Z1ZV

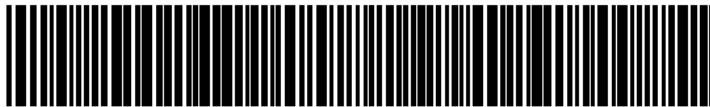
Rice Premium

Grade: **A**

Pack: **5 kg**

Net Weight: **5.000**

Batch: **B-2406**



Label generated from weighing ERP by punitinstrument.com

Large Label Select Max (5/8)

Punit Foods Pvt. Ltd.

22-06-2026
14:35

ops@punitfoods.in | +91 98765 43210

GST: 24AABCP1234Z1ZV

Wheat Flour

Grade: **A**

Pack: **10 kg**

Gross Weight: **10.250**

Tare Weight: **0.250**

Net Weight: **10.000**



Label generated from weighing ERP by punitinstrument.com

Extra Large Label Select Max (7/9)

Punit Foods Pvt. Ltd.

22-06-2026
14:35

ops@punitfoods.in | +91 98765 43210
GST: 24AABCP1234Z1ZV

Mixed Pulses

Grade:	A	Pack:	10 kg
Color:	Yellow	Origin:	Gujarat
Gross:	10.250	Tare:	0.250
Net:	10.000		



Label generated from weighing ERP by punitinstrument.com

Wholesale Pack

Punit Foods Pvt. Ltd.

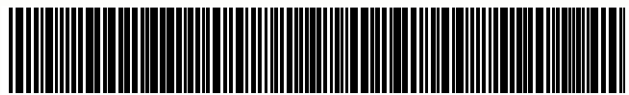
22-06-2026
14:35

ops@punitfoods.in | +91 98765 43210
GST: 24AABCP1234Z1ZV
On working day 11:00AM - 6:00PM

Wholesale Pack

Wholesale Spice Box

Sr No:	1
Grade:	Export
Pack:	25 kg
Gross Weight:	25.800



Small Seven (5)

Punit Foods Pvt. Ltd.

Spice Mix 7

22-06-2026

14:35

Sr No : 1

Color : Red

Size : Small

Units : 25

Net Weight : 1.250



DryFruit Label Format

22-06-2026 14:35

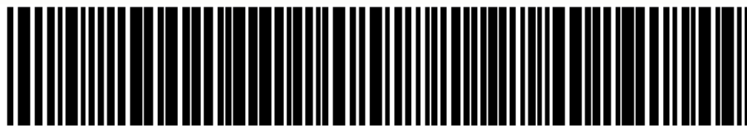
Premium Cashew W320

Grade : W320

Batch : DF-2406

Net Weight : 1.250 Kg

Units : 25



Custom Runtime Template Example

Custom 75x75 Template

company_name: Punit Foods

product_name: Besan Flour

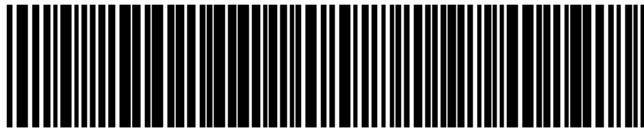
gross_weight: 1.500

tare_weight: 0.250

net_weight: 1.250

attr_grade: Grade: A

sr_no: 1



footer: Thank you

8. Example Print Payload

Field	Example
Product	Besan Flour
Barcode	A7K91
Attributes	Grade: A, Pack: 1 kg, Color: Yellow
Gross Weight	1.500
Tare Weight	0.250
Net Weight	1.250
Units	25, when unitConversion is enabled and unitValue x net weight equals 25
Serial Number	1, when Serial Number toggle is enabled